

JEE MAIN REVISION MASTER: ELASTICITY

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Hooke's Law & Young's Modulus of Wires	9 PYQs	Elastic Moduli
Standard	Elastic Potential Energy & Wire Elongation Due to Weight	1 PYQs	Energy & Special Cases

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Hooke's Law & Young's Modulus of Wires	$Y = \frac{F L}{A \Delta L} = \frac{M g L}{\pi r^2 \Delta L}$, $\Delta L \propto \frac{L}{r^2}$
Elastic Potential Energy & Wire Elongation Due to Weight	$U = \frac{1}{2} F \Delta L$, $u = \frac{1}{2} \text{Stress} \times \text{Strain}$, $\Delta L_{\text{self}} = \frac{\rho g L^2}{2 Y}$, $\sigma = \frac{\Delta d}{d} \Delta L / L$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hooke's Law & Young's Modulus of Wires (High Yield — 9 PYQs)

Core Principle: Hooke's law in 1D: Longitudinal stress $\sigma = (F / A)$ and strain $\epsilon = (\Delta L / L)$. Young's modulus $Y = (F L / A \Delta L) = (F L / \pi r^2 \Delta L)$.

■ **Caution / JEE Pitfall:** Ratio problems with diameter: remember $A \propto d^2$, so $\Delta L \propto (L / d^2)$. If diameter is doubled, elongation becomes 1/4!

■ **Speed Trick:** For two wires under equal tension: $(\Delta L_1 / \Delta L_2) = \left((L_1 / L_2) \right) \left((d_2 / d_1) \right)^2 \left((Y_2 / Y_1) \right)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the..." [Ans: D]
- [2025 Shift] Q#2 (MCQ): "Two wires A and B are made of same material. Their diameters are in the ratio of 1:2 and lengths are in the ratio of 1:3. If they are s..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "A thick metal wire of density ρ and length L is hung from a rigid support. The increase in length of the wire due to its own weight i..." [Ans: N/A]

3.2 Elastic Potential Energy & Wire Elongation Due to Weight (Standard Yield — 1 PYQs)

Core Principle: Work done in stretching a wire $W = U = \frac{1}{2} F \Delta L = \frac{1}{2} Y (\text{strain})^2 \times \text{Volume}$. Elongation of a hanging wire under its own weight $\Delta L = \left(\frac{\rho g L^2}{2 Y}\right)$. Poisson's ratio $\sigma = -\left(\frac{\Delta r / r}{\Delta L / L}\right)$.

■ **Caution / JEE Pitfall:** Elongation due to own weight has factor $1/2$ in denominator because effective center of gravity acts at $L/2$!

■ **Speed Trick:** Poisson's ratio theoretical limits: $-1 \leq \sigma \leq 0.5$. For practical materials, $0 \leq \sigma \leq 0.5$.

Representative Source PYQs:

- **[2023 Shift] Q#4 (MCQ):** "A stretched wire of a material whose young's modulus $Y = 2 \times 10^{11} \text{Nm}^{-2}$ has poisson's ratio 0.25 . Its lateral strain $\epsilon_{\text{...}}$ " **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

There are two wires of same material and same length while the diameter of second wire is two times the diameter of the first wire. Then the ratio of extensions produced in the wires by applying same load will be

- (A) 1:1
- (B) 1:2
- (C) 2:1
- (D) 4:1 **[Correct]**

Correct Answer: D

Solution Analysis: For a wire, the extension produced by a load is given by $\Delta L = (FL / AY)$ Here, F is the load applied, L is the length of the wire, A is the area of cross-section, Y is Young's modulus. Both wires have the same material, so Y is the same. Both have the same length, and the same load is applied. So, $\Delta L \propto (1 / A)$ Now, area of cross-section of a wire is $A = \frac{\pi d^2}{4}$ So area is proportional to the square of diameter: $A \propto d^2$ Given, $d_2 = 2d_1$ So, $A_2 = 4A_1$ Since extension is inversely proportional to area, $\frac{\Delta L_1}{\Delta L_2} = \frac{A_2}{A_1}$